

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			Indicative content AO1 • Temperature increases as acid is added up to 25 cm³ • pH decreases from pH 14-8 • As more alkali is neutralised • Maximum temperature at point when all alkali is just neutralised / pH is 7 • Additional acid results in no further reaction but temperature decreases (due to addition of cooler solution / returning to room temperature) • pH decreases from pH 6-1 / below 7 AO2 • Alkali is neutralised as first 25 cm³ of acid is added • 25 cm³ is point of complete neutralisation • 25-50 cm³ acid is in excess 5-6 marks Good understanding of the link between temperature and neutralisation; volumes from the graph used to support the explanation <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Full description of the temperature change; some understanding of the link between temperature and neutralisation <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Basic description of the temperature change shown on the graph <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>	3	3		6		6

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			$\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ ignore any state symbols and attempt to balance	1			1		
				Question 7 total	4	3	0	7	0	6